

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Only statistical tables provided by the institute are allowed. Electronic calculators are allowed. No other tables or formula sheets are allowed

1. The following are 9 independent observations from a continuous random variable X , that takes values in $\mathbf{R}_+$. The random variable has expectation $E(X) = m(\theta) = e^{\theta-1}$. The parameter space is $\Theta = [1, \infty)$.

0.653, 0.568, 0.107, 0.141, 0.857, 0.622, 0.431, 0.631, 0.735.

Find the least squares estimator of θ . [10p]

2. In an experiment with a radioactive material one measures the times between the individual decay events, with the use of a Geiger meter and a stop watch. It is known that the times between the decay events (the times for individual counts as registered by the Geiger meter) are independent and exponentially distributed. Let T_i denote the times between decays, so that $T_i, i = 1, \dots, 10$ are independent and exponentially distributed, and denote $\theta = E(T_i)$. The observations are

0.920, 0.0371, 0.117, 0.299, 1.16, 0.300, 0.130, 1.09, 2.79, 0.772.

- (a) Find the maximum likelihood estimator of θ . [3p]
- (b) Introduce $Z = \min(X_1, \dots, X_n)$ and find the plug-in estimator of $P(Z \leq 1)$, based on the estimator in (a). [3p]
- (c) Perform an exact test of

$$\begin{aligned} H_0 : & \quad \theta \leq \frac{1}{2}, \\ H_1 : & \quad \theta > \frac{1}{2}. \end{aligned}$$

[4p]

3. Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{1}{2\theta} & -\theta \leq x \leq 0, \\ \frac{1}{2}\theta(1-\theta)^{x-1} & x = 1, 2, 3, \dots, \end{cases}$$

with respect to the measure $\mu_1(x)1\{x \leq 0\} + \mu_2(x)1\{x > 0\}$, with μ_1 length measure and μ_2 counting measure. The parameter space is $\Theta = [0, 1]$. We have the following i.i.d. observations of X

−0.271, −0.243, −0.390, −0.336, −0.0214, −0.0602, −0.158, 3, 1, 2, 1, 1.

(a) Find the maximum likelihood estimator of θ . [5p]

(b) Find the plug-in estimator of $P(-1/4 \leq X < 3)$. [5p]

4. Suppose that X is $Bin(10, \theta)$ distributed and suppose that we have the following 5 i.i.d. observations of X

2, 0, 1, 0, 0.

(a) Find the maximum likelihood estimator of θ . [5p]

(b) Based on the data, perform an exact test of the hypotheses

$$H_0 : \theta \geq 1/3,$$

$$H_1 : \theta < 1/3.$$

on significance level 0.01. [5p]

(c) Suppose that you are given the following test: Reject the null hypothesis if the sum of the observations is strictly less than 5. What is the approximate power of this test, using an adequate approximation, at $\theta = 1/5$? [10p]

5. Suppose that one repeatedly makes temperature measurements of the water temperature at depths of 1 meter and at 10 meters in the waters near a power plant. The measurements may be assumed to follow the Normal (Gaussian) distribution, assuming the same variance at the two different depths but with possibly different expectations, and all measurements are assumed independent. The following are the data (in degree Celcius) from the measurements.

at 1 meter 21.0, 20.1, 21.3, 22.2

at 10 meters 21.2, 19.5, 19.8, 21.4, 20.0, 21.2, 19.6.

(a) Construct a 95% two-sided confidence interval for the expected temperature difference between temperatures at 1 and at 10 meters. [10p]

(b) Construct a 95 % lower bounded confidence interval for $1/\text{variance}$. [10p]

6. A medical research facility examines the effect of vitamin supplements for nicotine smokers. The reason is that nicotine seems to become more prolific among athletes, and the researchers want to find ways to mitigate the negative aspects of nicotine. For each individual i one measures the vitamin C consumption x_{1i} (mg/day), the vitamin D consumption x_{2i} (mg/day) and the amount of nicotine consumption x_{3i} (mg/day). It is known that nicotine spikes blood sugar, and therefore one uses as response variable y_i the blood sugar level as a measure of the overall adverse effect of nicotine on the body. The data is assumed to follow the linear model

$$y_i = \theta_0 + \theta_1(x_{1i} - \bar{x}_1) + \theta_2(x_{2i} - \bar{x}_2) + \theta_3(x_{3i} - \bar{x}_3) + \epsilon_i,$$

for $i = 1, \dots, 100$, with ϵ_i i.i.d. $N(0, \sigma^2)$ -distributed random variables, and $\sigma^2 = 3$ assumed to be known.

The summarised data are, on matrix form,

$$\begin{aligned}(X^T X)^{-1} &= \begin{pmatrix} 1 \cdot 10^{-2} & 0 & 0 & 0 \\ 0 & 1.397 \cdot 10^{-2} & -2.313 \cdot 10^{-4} & 9.140 \cdot 10^{-5} \\ 0 & -2.313 \cdot 10^{-4} & 7.474 \cdot 10^{-4} & -2.938 \cdot 10^{-5} \\ 0 & 9.140 \cdot 10^{-5} & -2.938 \cdot 10^{-5} & 8.687 \cdot 10^{-5} \end{pmatrix}, \\ X^T Y &= \begin{pmatrix} 796.4 \\ -126.9 \\ -22.77 \\ -36.99 \end{pmatrix}, \\ \bar{x}_{1.} &= 100.0, \\ \bar{x}_{2.} &= 149.3, \\ \bar{x}_{3.} &= 50.32.\end{aligned}$$

- (a) Construct a 95% two-sided confidence interval for the effect θ_1 of vitamin C. [6p]
- (b) Test if the effects of vitamin C and vitamin D are the same. [6p]
- (c) Construct a 95% two-sided confidence interval for the expected blood sugar level for an individual that consumes $x_1 = 100$ (*mg/day*) vitamin C, $x_2 = 50$ (*mg/day*) vitamin D and $x_3 = 100$ (*mg/day*) nicotine. [8p]